

ELEC 7410 Homework Assignment #6 Oct. 23, 2025

(6 problems, Due: Nov. 4, 2025)

Any late homework submission will incur 15% penalty per day, with any fraction of a day counted as full day.

1. (10 points) (**Linear Estimation of Random Vectors**) Problem 32, Chapter 8, Gubner.

32. Let $\hat{X}$ denote the linear MMSE estimate of the vector X based on the observation vector Y . Now suppose that $Z := BX$. Let $\hat{Z}$ denote the linear MMSE estimate of Z based on Y . Show that $\hat{Z} = B\hat{X}$.

2. (10 points) (**Nonlinear Estimation of Random Vectors**) Problem 46 (**MMSE only**), Chapter 8, Gubner.

46. If X and Y are positive random variables with joint density

$$f_{XY}(x, y) = (x/y^2)e^{-(x/y)^2/2} \cdot \lambda e^{-\lambda y}, \quad x, y > 0,$$

find ~~both~~ the MMSE and MAP estimators of X given Y (~~the estimators should be different~~).

3. (10 points) (**Strict-sense and Wide-sense Stationary Processes**) Problem 8, Chapter 10, Gubner.

8. Let X_t be a zero-mean, WSS process with correlation function $R_X(\tau)$. Let $Y_t := X_t \cos(2\pi ft + \Theta)$, where $\Theta \sim \text{uniform}[-\pi, \pi]$ and Θ is independent of the process X_t .

- (a) Find the correlation function of Y_t .
- (b) Find the cross-correlation function of X_t and Y_t .
- (c) Is Y_t WSS?

4. (10 points) (**Strict-sense and Wide-sense Stationary Processes**) Problem 14, Chapter 10, Gubner.

14. Let $q(t)$ have period T_0 , and let $T \sim \text{uniform}[0, T_0]$. Is $X_t := q(t + T)$ WSS? Justify your answer.

5. (10 points) (**The Poisson Process**) Problem 3, Chapter 11, Gubner.

3. Let N_t be a Poisson process with rate $\lambda = 2$, and consider a fixed observation interval $(0, 5]$.

- (a) What is the probability that $N_5 = 10$?
- (b) What is the probability that $N_i - N_{i-1} = 2$ for all $i = 1, \dots, 5$?

6. (10 points) (**The Poisson Process**) Problem 7, Chapter 11, Gubner.

7. Let N_t be a Poisson process with rate λ , and let $\Delta t > 0$.

- (a) Show that $N_{t+\Delta t} - N_t$ and N_t are independent.
- (b) Show that $P(N_{t+\Delta t} = k + \ell | N_t = k) = P(N_{t+\Delta t} - N_t = \ell)$.
- (c) Evaluate $P(N_t = k | N_{t+\Delta t} = k + \ell)$.
- (d) Show that as a function of $k = 0, \dots, n$, $P(N_t = k | N_{t+\Delta t} = n)$ has the binomial(n, p) probability mass function and identify p .